MESA SCHOOL OF BUSINESS

Data Foundations · Pre-Read

Session 4: Meesho

Big Mastery Question: Our top-selling product has our worst resellers. Prove it or kill the claim.

Before you come to class

Read this once, slowly. This session's big idea: your data almost never lives in ONE table. This time you have an Orders sheet and a separate Reseller sheet, and you need to pull information from one into the other. Every table below shows real column letters and row numbers — and since you'll be working across TWO separate sheets this session, pay close attention to which sheet each column letter belongs to.

The company: Meesho lets everyday people — 'resellers' — sell products online without holding any inventory themselves; Meesho handles the catalogue and shipping. Every order links a reseller to a product. Your job this session: figure out whether Meesho's best-selling product is actually being sold badly by its worst resellers — and that means pulling reseller information into the orders table.

0. The Two Datasets You'll Be Working With

Sheet 1 — "Orders" (7 columns: order_id, date, reseller_id, product_id, units, order_value, status). A 5-column slice:

	A	C	D	E	F
1	order_id	reseller_id	product_id	units	order_value
2	M4001	R501	P100	3	900
3	M4002	R502	P101	1	250
4	M4003	R501	P100	2	600
5	M4004	R503	P102	5	1500
6	M4005	R502	P101	2	500
7	M4006	R501	P103	1	300

On the Orders sheet: A = order_id, C = reseller_id, D = product_id, E = units, F = order_value. (B = date, G = status, not shown.)

Sheet 2 — "Resellers" (a completely separate tab, 5 columns: reseller_id, name, city, tier, joined_date). A 4-column slice:

	A	B	C	D
1	reseller_id	name	city	tier
2	R501	Priya Textiles	Surat	Gold
3	R502	Anand Fashions	Jaipur	Silver

4	R503	Meera Crafts	Indore	Gold

On the Resellers sheet, column A starts fresh again — reseller_id is A, name is B, city is C, tier is D. This is a different tab, so it has its own column A, B, C...

1. The Idea of a Key & a Join

Real world reference:

Imagine two friends each holding half a jigsaw puzzle. The only way to combine them into one picture is finding the matching edge piece both halves share. reseller_id is exactly that matching edge — it's the ONE column that exists identically in both the Orders sheet and the Resellers sheet, and it's what lets you connect them.

In Meesho's world:

The Orders sheet knows WHAT was sold and by whom (via reseller_id) — but nothing about that reseller. The Resellers sheet knows everything about each reseller — but nothing about what they sold. Neither sheet alone can answer "which resellers sell our top product badly?" You need to JOIN them — pull reseller details across into the Orders sheet, matched by reseller_id.

Why this matters before any formula:

Every lookup function you learn this session — VLOOKUP, XLOOKUP, INDEX+MATCH — does the exact same job: given a KEY (reseller_id), find the matching row in another table, and pull back one piece of information from it. Once you see the join, the formulas are just different ways of asking the same question.

2. VLOOKUP

Real world reference:

Imagine a school register: you know a student's roll number, and you scan down the ROLL NUMBER column until you find a match, then read ACROSS that same row to find their name. VLOOKUP does exactly that — V for "Vertical": it searches down the first column of a range, and reads across to pull back a value.

In Meesho's world:

For every order, you know the reseller_id ("R501"). VLOOKUP takes that ID, searches down column A of the Resellers sheet until it finds a match, then reads across to column B and pulls back the reseller's name.

Syntax:

```
=VLOOKUP(search_key, range, column_index, is_sorted)  Example: =VLOOKUP(C2, Resellers!A:D, 2, FALSE) (column_index 2 = the 2nd column of the range, counting from A = column 1)
```

See it in the table:

	A	C	H
1	**order_id**	**reseller_id**	**Reseller Name**
2	M4001	R501	=VLOOKUP(C2,Resellers!A:D,2,FALSE) → Priya Textiles
3	M4002	R502	=VLOOKUP(C3,Resellers!A:D,2,FALSE) → Anand Fashions

4	M4004	R503	=VLOOKUP(C5,Resellers!A:D,2,FALSE) → Meera Crafts

FALSE at the end means "exact match only" — almost always what you want. Leave it out (or use TRUE) and Sheets may return the wrong reseller entirely.

3. XLOOKUP

Real world reference:

VLOOKUP can only look RIGHT of the key column, and always counts columns by number, which breaks if someone inserts a new column. XLOOKUP is the newer, more flexible cousin — it can look in any direction, and instead of counting columns, you just point directly at the column you want.

In Meesho's world:

Same task — pull the reseller's city into the Orders sheet — but this time using XLOOKUP, AND handling the case where a reseller_id doesn't exist in the Resellers sheet at all (a data-entry mistake), by showing "Not Found" instead of an ugly error.

Syntax:

```
=XLOOKUP(search_key, lookup_range, result_range, [if_not_found])  Example:
=XLOOKUP(C2, Resellers!A:A, Resellers!C:C, "Not Found")
```

See it in the table:

	A	C	H
1	**order_id**	**reseller_id**	**Reseller City**
2	M4001	R501	=XLOOKUP(C2,Resellers!A:A,Resellers!C:C,"Not Found") → Surat
3	M4002	R502	=XLOOKUP(C3,Resellers!A:A,Resellers!C:C,"Not Found") → Jaipur
7	M4006	R999	=XLOOKUP(C7,Resellers!A:A,Resellers!C:C,"Not Found") → Not Found

R999 doesn't exist in the Resellers sheet — VLOOKUP would throw a #N/A error here, but XLOOKUP's if_not_found argument catches it cleanly.

4. HLOOKUP

Real world reference:

Same idea as VLOOKUP, just rotated 90 degrees. Instead of scanning DOWN a column and reading ACROSS, HLOOKUP scans ACROSS a row and reads DOWN. H is for "Horizontal." It's rarer, because most real data is organised in columns, not rows — but tables like commission grids are often laid out horizontally.

In Meesho's world:

Meesho's commission rates are stored in a small horizontal reference table — tier names running ACROSS row 1, and commission percentages running across row 2 directly below them. HLOOKUP scans across row 1 for a matching tier, then reads down to row 2 to get the rate.

Syntax:

```
=HLOOKUP(search_key, range, row_index, is_sorted)  Example: =HLOOKUP("Gold", B1:D2, 2, FALSE)
```

See it in the table:

	A	B	C	D
1	**Metric**	**Bronze**	**Silver**	**Gold**
2	Commission %	5%	8%	12%

=HLOOKUP("Gold", B1:D2, 2, FALSE) scans row 1 for "Gold", finds it in column D, then reads DOWN to row 2 of that same column → 12%.

5. INDEX

Real world reference:

Imagine a bingo grid, and someone tells you "the answer is in row 3, column 2." You don't search for anything — you just go directly to that position and read what's there. That's exactly what INDEX does: give it a position, it hands back the value sitting at that spot.

In Meesho's world:

If you already know a reseller is the 3rd row in the Resellers table, INDEX can pull back any of their details instantly — no searching required, just "go to row 3, column 3" to get their city.

Syntax:

```
=INDEX(range, row_number, column_number)  Example: =INDEX(A2:D4, 3, 3)
```

See it in the table:

	A	B	C	D
1	**reseller_id**	**name**	**city**	**tier**
2	R501	Priya Textiles	Surat	Gold
3	R502	Anand Fashions	Jaipur	Silver
4	R503	Meera Crafts	Indore	Gold

=INDEX(A2:D4, 3, 3) → counts 3 rows down and 3 columns across WITHIN the range A2:D4, landing on "Indore". Row and column numbers are relative to the range, not the whole sheet.

6. MATCH

Real world reference:

If INDEX is "go directly to position 3," MATCH is the opposite question: "what position is R502 sitting at?" It doesn't return the value itself — it returns the POSITION where it was found.

In Meesho's world:

You want to know where R502 sits in the Resellers list — not to read it yourself, but because that position number is exactly what INDEX needs as its row_number input. MATCH and INDEX are built to work together.

Syntax:

```
=MATCH(search_key, range, 0)  Example: =MATCH("R502", A2:A4, 0) (the 0 means exact match)
```

See it in the table:

	A
1	**reseller_id**
2	R501
3	R502
4	R503
5	**=MATCH("R502",A2:A4,0) = 2**

R502 is the 2nd row WITHIN the range A2:A4 (not the 3rd row of the whole sheet) — MATCH counts relative to where the range starts, just like INDEX does.

7. INDEX + MATCH

Real world reference:

Put the two together and you get something powerful: MATCH finds the POSITION of what you're looking for, and INDEX uses that position to fetch the value — all in one formula. It's VLOOKUP's job, done in two steps welded together, and it fixes VLOOKUP's biggest weakness: it can look in ANY direction, not just rightward.

In Meesho's world:

Pull a reseller's name into the Orders sheet — the exact same result as the VLOOKUP example earlier — but built with INDEX+MATCH instead. The advantage: if someone later inserts a new column into the Resellers sheet, this formula keeps working. A plain VLOOKUP with a hardcoded column number would silently break.

Syntax:

```
=INDEX(return_range, MATCH(search_key, lookup_range, 0))  Example: =INDEX(Resellers!B2:B4, MATCH(C2, Resellers!A2:A4, 0))
```

See it in the table:

	A	C	H
1	**order_id**	**reseller_id**	**Reseller Name**
2	M4001	R501	=INDEX(Resellers!B2:B4,MATCH(C2,Resellers!A2:A4,0)) → Priya Textiles
3	M4002	R502	=INDEX(Resellers!B2:B4,MATCH(C3,Resellers!A2:A4,0)) → Anand Fashions

MATCH finds WHERE "R501" sits inside Resellers!A2:A4 (position 1). INDEX then goes to that same position in Resellers!B2:B4 and reads "Priya Textiles". Same final answer as VLOOKUP — different, sturdier route.

Coming to class

You now know how to connect two separate tables using a shared key, four different ways. In class, we'll pull reseller details onto the full Orders sheet and find out whether Meesho's best-selling product really is being let down by its worst resellers.